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European forest disturbance alerting using Sentinel-1☆

Sietse van der Woude [a,*], Johannes Reiche [a,*], Johannes Balling [a], Gert-Jan Nabuurs [b,c], Frank Sterck [b], Anne-Juul Welsink [a], Bart Slagter [a], Martin Herold [a,d]

[a] Wageningen University & Research, Laboratory of Geo-information Science and Remote Sensing, Droevendaalsesteeg 3, 6708 PB Wageningen, the Netherlands
[b] Forest Ecology and Forest Management, Wageningen University and Research, Droevendaalsesteeg 3, 6708 PB Wageningen, the Netherlands
[c] Wageningen Environmental Research, Droevendaalsesteeg 3, 6708 PB Wageningen, the Netherlands
[d] GFZ German Research Centre for Geosciences, Remote Sensing and Geoinformatics Section, Telegrafenberg, 14473 Potsdam, Germany

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ABSTRACT

We demonstrate a near-real time forest disturbance alerting system for Europe using Sentinel-1 radar data. Sentinel-1 radar can penetrate clouds and offers high spatial (~20 m) and temporal (3- to 6-day) detail. We directly integrated near-real time ERA5-Land temperature and Copernicus forest type data into the disturbance detection framework to address freezing temperatures and seasonal phenology, both of which influence the Sentinel-1 backscatter signal and thus need to be accounted for. This facilitates year-round monitoring across a range of environmental conditions (sub-zero, wet and dry) and forest types (coniferous, deciduous) throughout the boreal, temperate, and Mediterranean forests of Europe. Validation across Europe showed high accuracy, with a user accuracy of 91.2% (±1.3%) and producer accuracy of 74.5% (±6.0%). User accuracy increased to 99% (±0.4%) when excluding errors in the European-scale forest cover mask primarily caused by local overestimation of forest height and density. Disturbances were detected with a median delay of 27 days relative to the first high-resolution optical Planet reference image, which can further be reduced to 1 day through retrospective event-based correction of late detection bias. Compared to existing annual optical-based products, our method improves the detection of small-scale disturbances such as group fellings in Romania. We generated European-scale estimates of intra-annual disturbance seasonality, capturing variation in forest management practices and disturbance regimes such as winter harvesting in northern Europe, spring sanitation cutting in central Europe, and summer wildfires in southern Europe. Overall, this alerting system provides timely and detailed forest disturbance information in support of sustainable forest management, biodiversity conservation, carbon accounting, and law enforcement efforts across Europe. The alerts are available at https://wurnrt-raddeurope.projects.earthengine.app/view/radd-europe.

1. Introduction

Accurate and timely estimates of where and when forest across Europe are being disturbed are important for forest management, law enforcement, carbon accounting, and biodiversity conservation. Forests occupy an increasingly central role in achieving sustainable policy goals such as those outlined in the 2030 Forest Strategy (European Commission, 2021a). These policies promote wood as a sustainable building material and as a fuel source for biomass power plants, both of which are expected to increase the harvesting pressure on forests (Bell et al., 2018). Simultaneously, forests are expected to contribute to reaching biodiversity targets, such as described in the European 2030 Biodiversity Strategy (European Commission, 2021b) and to reaching the climate mitigation targets set in the certification framework for carbon removals (European Commission, 2022). These policies have resulted in a growing need for consistent and timely information on the state of the forests to balance the multiple demands imposed upon them.

This need has been fueled by the realization that forests are at risk, potentially impeding the realization of European policy goals related to climate and nature restoration targets. These risks are linked to slow-moving processes such as climate change-related droughts and excess nutrient deposition (FOREST EUROPE, 2020), as well as sudden events leading to a loss of forest cover and biomass. These include forest fires, windthrows, and pest infestations (Patacca et al., 2023). Studies have

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* Corresponding authors.
E-mail addresses: sietse.vanderwoude@wur.nl (S. van der Woude), Johannes.reiche@wur.nl (J. Reiche).

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estimated that canopy mortality has doubled in European forests between 1984 and 2016 (Senf et al., 2018), while the spatial dispersion of disturbances has become increasingly scattered (Senf and Seidl, 2018). As a result, traditional sources of information such as the sample-based 5- to 10-year ground-based forest inventory data cycles alone can no longer meet increasing forest monitoring demands.

Satellite-based optical and radar remote sensing offers extensive and consistent observations and has become key in estimating the extent and severity of forest disturbances. Large-scale forest disturbance products such as Global Forest Change (Hansen et al., 2013), European Forest Disturbance Atlas (EFDA) (Viana-Soto and Senf, 2025), and Tree Canopy Height Change (Turubanova et al., 2023) have been developed using 30 m resolution Landsat data and provide disturbance area estimates for multi-decadal time spans. However, they incur a minimum one-year data delivery lag due to their reliance on cloud-free observations, limiting their application for rapid response applications such as law enforcement and emergency management after windstorms or fires.

In contrast, synthetic aperture radar (SAR) offers year-round observations unaffected by cloud cover and illumination conditions. The Sentinel-1C-band SAR satellite constellation has increased the availability of high-resolution (~20 m) radar data since its launch in 2014, offering revisit intervals of up to 6 days at the equator and up to 3 days at higher latitudes in Europe, along with open data distribution (European Space Agency, 2026). This has enabled the development of operational, near-real time forest disturbance detection systems in the tropics such as the pan-tropical Radar for Detection of Deforestation (RADD) alerts (Reiche et al., 2021), TropiSCO (Mermoz et al., 2021) and the Real Time Deforestation Detection System (DETER-R) (Doblas et al., 2022). These SAR-based systems have proven to be highly complementary to optical near-real time detection systems (Doblas et al., 2023; Reiche et al., 2024).

Despite successful implementation in the tropics, the use of C-band SAR backscatter data for detecting disturbances in temperate and boreal zones is limited by its strong sensitivity to freezing conditions (Benninga et al., 2019; Cohen et al., 2021; Lemmetyinen et al., 2022), and to a lesser degree, shifts in seasonal leaf phenology (Ahern et al., 1993; Frison et al., 2018; Rüetschi et al., 2017; Soudani et al., 2021). The sensitivity to freezing temperatures is primarily linked to changes in moisture content: freezing conditions decrease the liquid moisture content in the canopy, thereby reducing the dielectric constant of the scatterers. This reduces the interaction with the radar waves, and decreases backscatter (Rignot et al., 1994). Other related factors can also decrease backscatter, such as frost protection mechanisms in needle-leaf trees (Ahern et al., 1993; Chang et al., 2021), signal penetration to frozen ground in sparse canopies (Cohen et al., 2024, 2021), and wet snow cover (Koskinen et al., 2010; Naderpour et al., 2022).

The sensitivity of C-band SAR to shifts in seasonal leaf phenology is related to changes in structure: In leaf-on conditions, leaves are the dominant scatterers and mask woody components like twigs and branches. During transition to leaf-off conditions, these become increasingly exposed and act as volume scatterers, increasing cross-polarized backscatter (Ahern et al., 1993; Rüetschi et al., 2017). Seasonal leaf phenology is largely dependent on temperature and varies annually. High spring temperatures can induce early leaf senescence (Vitasse et al., 2009a, 2009b), thereby decreasing volume scattering and reducing backscatter (Frison et al., 2018; Rüetschi et al., 2017; Soudani et al., 2021).

These factors must be accounted for to avoid confusion between seasonal variation and actual forest disturbances. Signal-based inference methods such as harmonic modelling (Shimizu et al., 2019), cumulative sum (Ruiz-Ramos et al., 2020), and Kalman filtering (Puhm et al., 2020) have been applied in Sentinel-1-based forest monitoring for this purpose. These methods are limited because they model the seasonal patterns in the backscatter timeseries itself, rather than the environmental conditions driving these patterns. They struggle to capture shifts in seasonality and stochastic weather fluctuations at daily and weekly time scales (Reiche et al., 2018a). Signal-based inference alone remains highly uncertain unless external covariates are incorporated, especially considering increases in weather extremes due to climate change (Cassou and Cattiaux, 2016; Menzel et al., 2006). Previous efforts have been made to incorporate climatic data, but these have been limited to filtering out observations affected by freezing temperatures (Mullissa et al., 2024) and snow cover (Mermoz et al., 2024).

In this study, we expand our near-real time Sentinel-1 forest disturbance detection method originally developed for the humid tropics (Reiche et al., 2021) to the seasonal boreal, temperate, and Mediterranean forests of Europe. By directly integrating temperature data from ERA5-Land (Muñoz-Sabater et al., 2021) and Copernicus forest type data, we explicitly account for the effect of freezing conditions and seasonal shifts in leaf phenology on the Sentinel-1 backscatter signal. We demonstrate this approach by generating spatially detailed (10 m × 10 m) weekly forest disturbance alerts over a four-year period (2020–2023). We evaluate their spatial accuracy and timeliness and compare our results with currently available optically-based European disturbance products (Hansen et al., 2013; Viana-Soto and Senf, 2025). Furthermore, we analyze seasonal disturbance regimes across Europe to highlight the benefits of intra-annual disturbance detection.

2. Study area and definitions

2.1. Study area

Our study area covered all EU member states except Cyprus and overseas territories, and included the following non-EU countries: Albania, Andorra, Bosnia and Herzegovina, Guernsey, Isle of Man, Jersey, Kosovo, Liechtenstein, Monaco, Montenegro, North Macedonia, Norway, San Marino, Serbia, Switzerland and United Kingdom. We defined four geographic regions for statistical analysis: North, East, South, and West Europe (Fig. 1).

2.2. Definitions

We created a forest cover mask for 2019 to restrict disturbance

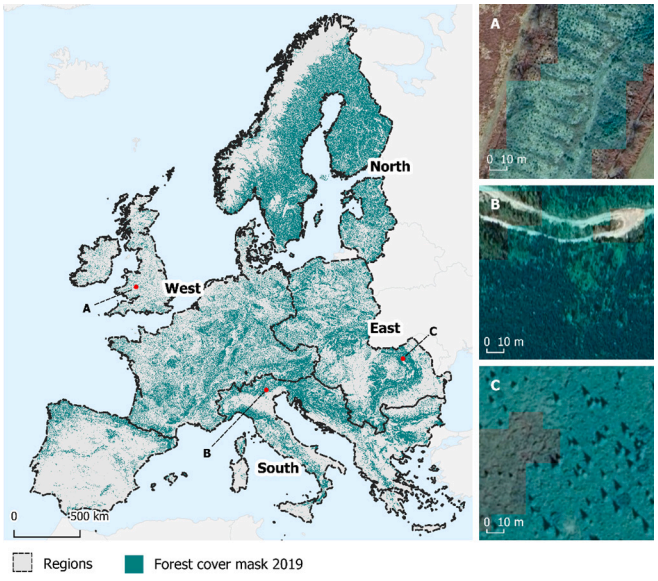


Fig. 1. Forest cover mask for 2019 with subdivision into geographic regions (North, East, South, and West Europe). Maps A-C show three examples of local inaccuracies in the forest cover mask (Google imagery used as basemap): (A) pre-2020 canopy cover removals in Wales (52.577311° N, −3.656523° E), (B) overestimation of forest height in Northern Italy (45.863691° N, 11.278484° E), (C) overestimation of forest density in sparse forest in the Romanian Carpathians (47.116965° N, 25.298427° E).

detections to undisturbed forest present at the beginning of the monitoring period in 2020 (Fig. 1). We defined forest as areas of non-agricultural and non-urban canopy cover of at least 50% with a minimum canopy height of 6 m. The forest cover mask for 2019 was derived by (i) intersecting the Copernicus Forest Type and Tree Cover products for 2018 (Buchhorn et al., 2020) and (ii) excluding areas with canopy removals before 2020 and canopy height < 6 m (Turubanova et al., 2023). 6 m canopy height was used rather than the more commonly used 5 m due to the lower accuracy of canopy cover extent (overestimation of canopy height) in the underlying product below 6 m (Turubanova et al., 2023). We further excluded remaining forest fragments smaller than 0.1 ha.

The continental scale of the input datasets (Buchhorn et al., 2020; Turubanova et al., 2023) occasionally resulted in local inaccuracies. This included inclusion of areas where forest cover had already been removed prior to 2020 (Fig. 1A), overestimation of forest height (Fig. 1B), and overestimation of forest density (Fig. 1C). Common to Landsat-based forest change products, some pre-2020 disturbance events were not captured by the algorithm used to generate the canopy removal data, primarily due to a lack of cloud-free Landsat data at the end of 2019 (Verhelst et al., 2021).

Forest disturbance was defined as removal of at least 50% canopy cover within a 10 × 10 m Sentinel-1 pixel (~0.01 ha), consistent with definitions of stand-replacing disturbances in other forest disturbance products (Hansen et al., 2013; Reiche et al., 2021; Senf and Seidl, 2021). Non-stand-replacing disturbances, such as individual tree breakage, defoliation or low-intensity thinning were not taken into consideration. We did not distinguish between disturbance agents, such as harvesting, pests, fires and storms.

3. Data

3.1. Sentinel-1

We used dual polarization (VV and VH) Sentinel-1 Ground Range Detected (GRD) products acquired in Interferometric Wide (IW) swath mode, as available in the Google Earth Engine catalog (Gorelick et al., 2017). Data from both ascending and descending orbits were used. The revisit interval of the available data was approximately 3 days with two operational satellites (Sentinel-1A and -1B; pre-December 2021), and approximately 6 days with a single satellite (Sentinel-1A after the failure of Sentinel-1B, post-December 2021). The imagery is multi-looked using a 5 × 1 window resulting in a resolution of approximately 20 m × 22 m and is provided at a 10 m pixel spacing to preserve maximum spatial detail. The pre-processing performed by Google include thermal noise, GRD border noise correction, radiometric calibration, and range-Doppler terrain correction (Google Developers, 2026).

We applied additional pre-processing steps to the Sentinel-1 GRD imagery. First, we performed radiometric slope correction using the Copernicus GLO-30 Digital Elevation Model (DEM) (Airbus, 2022). Second, we applied supplementary border noise corrections as described in Mullissa et al. (2021) to obtain gamma nought backscatter values, followed by conversion to decibels. Third, we applied incidence angle-specific masking to address variations in backscatter values caused by overlapping orbit paths. This was done by selecting the relative orbit path per pixel and per orbit direction in which the incidence angle per pixel was closest to the average incidence angle (~37.9°) of an image. This resulted in a decrease in effective revisit interval (6 days with two operational satellites and 12 days with one), but it ensured consistent backscatter values and temporal coverage independent of latitude. We did not apply additional speckle filtering in order to preserve the greatest possible spatial detail in the imagery (Bottani et al., 2024). This offers advantages in the detection of small disturbances and disturbances with irregular or narrow shapes such as roads, but may also introduce commission errors.

We calculated Grey Level Co-Occurrence Matrix (GLCM) Sum Average (Savg) as additional Sentinel-1 features (VV_{Savg} and VH_{Savg}) besides the VV and VH backscatter (VV_{BS} and VH_{BS}) using a 5-pixel window size (Balling et al., 2023; Hall-Beyer, 2017; Haralick et al., 1973). Compared to backscatter, GLCM Savg has demonstrated increased detection sensitivity for fire events and large-scale clearings with remaining structure such as stumps and slash (Balling et al., 2023; Gibson et al., 2023).

3.2. ERA5-Land

We employed ERA5-Land 2 m modelled surface air temperature data available in the Google Earth Engine catalog (Gorelick et al., 2017). This dataset provides near-real time global coverage (with ~1 week delay) at a spatial resolution of ~11 km (Muñoz-Sabater et al., 2021). We performed two pre-processing steps: (i) we addressed missing temperature data in coastal areas by buffering and extrapolating edge pixels using a 4 × 4-pixel gaussian kernel and (ii) we modified temperatures using the Copernicus GLO-30 DEM to account for the general decrease in air temperature with elevation. We used lapse rates linearly interpolated between 5 °C/km on December 31st and 6.5 °C/km on June 30th. These are consistent with observed seasonal patterns where greater lapse rates occur during warmer months (Rolland, 2003).

4. Methods

4.1. European forest disturbance alerting

Our forest disturbance detection methodology is based on an approach originally developed for the tropics (Reiche et al., 2015b, 2018a, 2021). A new forest disturbance is flagged by comparing new Sentinel-1 observations with historical stable forest metrics. Subsequent images are used to increase confidence, resulting in low- or high-confidence alerts. We modified this method for European forest conditions by directly integrating Sentinel-1 data with ERA5-Land temperature data to account for the effects of freezing temperatures and with Copernicus forest type data (Buchhorn et al., 2020) to address seasonal variations in leaf phenology (Sections 3.1.1 and 3.1.2). We validated the spatial accuracy and detection timeliness (Sections 3.2.1 and 3.2.2) of high-confidence alerts and compared detected disturbance area with annual optically-based disturbance products (Section 3.3). Finally, we performed an Europe-wide analysis of disturbance seasonality, for which we applied event-based bias correction as additional processing step retrospectively to improve detection timeliness (Section 3.4).

4.1.1. Temperature-based forest metrics

We calculated pixel-based historical temperature-based metrics to describe the derived distribution of stable forest (F) and non-forest (NF) backscatter within temperature intervals for each orbit direction separately. This was performed in five steps. First, for the three year historical period from 2017 to 2019, 800 Sentinel-1 images were sampled per 2° x 2° tile (or fewer, depending on availability), resulting in 138 to 342 observations per pixel. Second, VV_{BS} and VH_{BS} and VV_{Savg} and VH_{Savg} observations were temporally matched to ERA5-Land temperature data and grouped into three temperature intervals (i1-i3): frozen (<−5 °C), transitional (−5 °C to 5 °C), and non-frozen (>5 °C). This accounts for lower values in the frozen interval and higher values in the non-frozen interval in coniferous forest pixels, and the inverse in deciduous forest pixels (Ahern et al., 1993; Rüetschi et al., 2017). The transitional interval captures the high uncertainty in backscatter response around 0 °C, which arises from variations in snow cover dynamics (Cohen et al., 2024, 2021; Lemmetyinen et al., 2022), variations in leaf emergence and falling (Vitasse et al., 2009b, 2009a), and uncertainty in the modelled temperature data itself (Muñoz-Sabater et al., 2021). Third, we defined the derived F distribution per temperature interval as a gaussian distribution $N_i(\widetilde{x_i}, 2\sigma_i)$, where $\widetilde{x_i}$ denotes the expectation (median) and $2\sigma_i$ the standard deviation at temperature

interval *i*. Fourth, we defined the derived NF distribution as a gaussian distribution $N_i(\widetilde{x_i}\text{-}4\sigma_i, 2\sigma_i)$, where we assumed that the $\widetilde{x_i}$ of the NF distribution would be $4\sigma_i$ lower than the $\widetilde{x_i}$ of F. Although this general assumption was originally made for wet and dry tropical forests (Reiche et al., 2018b), we found that this was applicable to European forests as well (Appendix A).

4.1.2. Disturbance detection

Our disturbance detection method consisted of five steps (I-V) and was applied to each newly acquired Sentinel-1 backscatter image

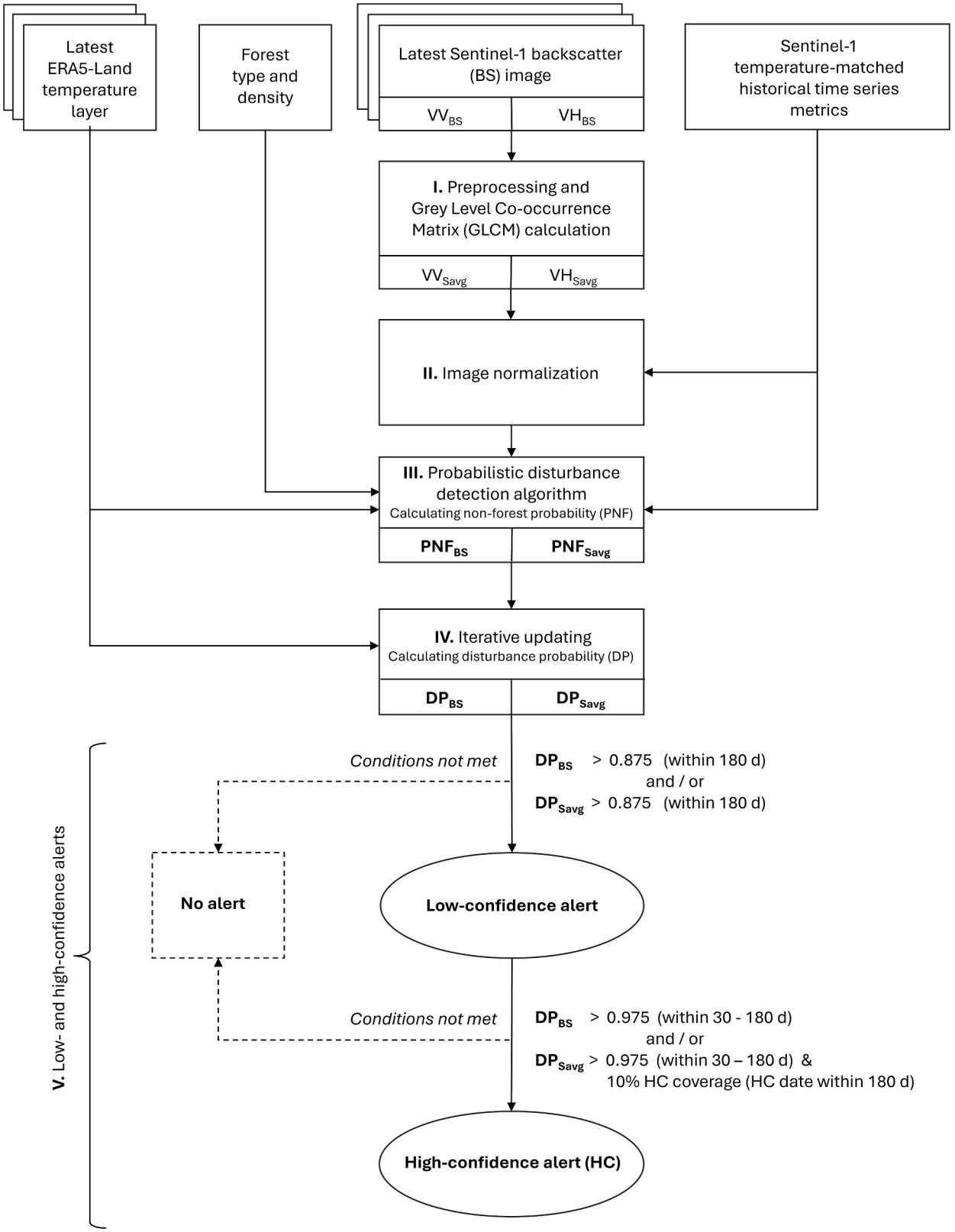


Fig. 2. Flowchart depicting the Sentinel-1-based forest disturbance alerting methodology. The roman numerals correspond to the steps described in Section 3.1.2.

(Fig. 2).

I. Preprocessing and GLCM calculation

Preprocessing (border noise, incidence angle-specific masking and slope correction; Mullissa et al., 2021) was applied to each dual-polarized backscatter (BS) image (VV_{BS} and VH_{BS}) followed by the calculation of GLCM Savg (VV_{Savg} and VH_{Savg}) features.

II. Image normalization

Image normalization was performed to mitigate local backscatter changes caused by climatic variability and the different responses of forest type and densities (Hamunyela et al., 2016; Reiche et al., 2018a). Precipitation and temperature can vary locally due to factors like elevation differences which are not captured adequately by the low resolution of the ERA5 temperature data. We clustered all backscatter and GLCM observations within the image using matching temperature intervals (frozen, transitional, non-frozen), forest types (deciduous and coniferous) and forest densities (high; ≥80% and low; <80%). For each cluster we calculated the mean of the historic medians (Section 3.1.1) and the mean of the current values. If the mean of current values at that timestep was lower than the mean of the historic values, we added the difference to each pixel value, resulting in normalized backscatter and GLCM values.

III. Probabilistic disturbance detection algorithm

New forest disturbances were detected using a probabilistic disturbance detection algorithm (Reiche et al., 2018b). The normalized backscatter (VV_{BS} and VH_{BS}) and GLCM Savg (VV_{Savg} and VH_{Savg}) observations (collectively denoted as S) were converted separately into conditional forest and non-forest probabilities $p(S_i|F_i)$ and $p(S_i|NF_i)$ at temperature interval i. This was done by extracting the coinciding observed ERA5 temperature and applying the corresponding temperature-specific F and NF probability density functions derived from observations in the historic period (Section 3.1.1). Then, conditional non-forest probabilities (PNF) were calculated used Bayes theorem:

$$PNF = P(NF_i S_i) = \frac{p(S_i NF_i)}{p(S_i NF_i) + p(S_i F_i)} \quad (1)$$

Similarly to Reiche et al. (2018b), we employed a block weighting function (0.05, 0.95) to remove extreme probabilities. The highest non-forest probability was selected for the backscatter (PNF_{BS}) and GLCM Savg observations (PNF_{Savg}) separately, and potential alerts were flagged when a threshold of 0.6 was exceeded for PNF_{BS} or PNF_{Savg}.

IV. Iterative updating of disturbance probabilities

Iterative Bayesian updating (Reiche et al., 2015a) was initiated for flagged alerts to calculate the disturbance probability (DP) separately for backscatter (DP_{BS}) and GLCM Savg (DP_{Savg}) using the past (t-1), current (t), and future observations ($t + j$) of non-forest probability (PNF_{BS} or PNF_{Savg}):

$$DP = P(D_t \quad PNF_{t+j}) = \frac{P(PNF_{t+j} D) P(D_t PNF_{t+j-1})}{P(PNF_{t+j})} \quad (2)$$

Here DP or $P(D_t|PNF_{t+j})$ denotes the posterior, representing the probability of disturbance at time t (D_t), given new evidence PNF_{t+j}. $P(PNF_{t+j})$ represents the total probability of PNF. j indexes the future observations ($j = 0, \ldots, n$). At the first timestep ($j = 0$), DP $P(D_t|PNF_t)$ was calculated using the PNF at the current step PNF_t as new evidence, and PNF_{t-1} as the prior. Afterwards, DP $P(D_t|PNF_{t+j})$ was updated at each timestep using incoming PNF values (PNF_{t+j}) as new evidence, and the previous iteration's DP as prior $P(D_t|PNF_{t+j-1})$. PNF values from both orbit directions were used to update DP, except where doing so would have halted iterative updating. In that case, the observation was ignored. Iterative updating (Eq. (2)) was halted if DP dropped below 0.5 or when high-confidence alerts were issued.

V. Low- and high confidence alerts

Low- and high confidence alerts were issued by applying pre-defined thresholds to disturbance probabilities DP_{BS} and DP_{Savg} with spatial and temporal constraints. An inverse buffer of half the GLCM kernel size (3 pixels) was applied to DP_{Savg} to remove overestimation of patch edges (Balling et al., 2023). Low-confidence alerts were issued when DP_{BS} or DP_{Savg} exceeded 0.875 within 180 days. High-confidence alerts were issued when the DP_{BS} exceeded 0.975 within a 30 to 180 day window. The 30-day minimum confirmation period was introduced in order to mitigate commissions related to temporary flooding and snowmelt events in low-density forest, which can incur sharp changes in the backscatter signal that endure longer than daily fluctuations induced by temperature (DeVries et al., 2020; Koskinen et al., 2010). High-confidence alerts were also issued when the DP_{Savg} exceeded 0.975 within a 30 to 180 day window, where at least 10% of the change patch overlapped with existing high-confidence (DP_{BS}-derived) alerts of which the mean date was within 180 days. This was done to reduce commission error during summer and autumn in deciduous forests of southern Europe to mitigate opposing geometric effects (shadow, layover) on backscatter at disturbance patch edges (van der Woude et al., 2024).

High-confidence detection dates were filtered using two subsequent spatial filters: (i) a 3 × 3-pixel modal kernel in order to smooth radar speckle-related patterns; and (ii) a 3 × 3-pixel plus-shaped kernel to smooth patch edges, removing pixels with only one direct neighbor. Finally, we applied a minimum mapping unit, removing patches smaller than 0.1 ha (10 4-connected pixels).

4.1.2.1. Detection example. Fig. 3 depicts an example of the disturbance detection methodology in Northern Sweden, where heavy snow cover and freezing temperatures are commonplace. Drops in backscatter over stable forest during freezing temperatures are visible, as well as the high variance of backscatter in the transitional temperature interval during snowmelt and snow accumulation. In point B, a change was flagged while in the transitional interval, since the low backscatter values resulted in high PNF values. After several observations in which DP was iteratively updated, the 0.975 threshold was exceeded and the change was identified as a high-confidence alert. Point A was correctly identified as non-disturbed, since the backscatter values were more likely within the range of the temperature-specific derived F distributions, rather than the derived NF distributions.

4.2. Validation

4.2.1. Assessing spatial accuracy

We validated the European Sentinel-1-based forest disturbance high-confidence alert product for a 4-year period (2020-2023) in a near-real time monitoring scenario through visual inspection using monthly composites of very-high-resolution multispectral PlanetScope data (~4.7 m) (Boshuizen et al., 2014). The alerting method was initiated in January 2019, but alerts occurring before 2020 were removed to avoid attributing erroneously unaccounted disturbances in the 2019 forest cover mask to early 2020 (Section 2.1, Fig. 1). We continued disturbance detection through June 2024 in order to allow low-confidence alerts detected near the end of the validation period (2023) to reach high confidence, but we removed new alerts after 2023.

A probability sampling design was applied (Stehman et al., 2003) using two stable forest and two forest disturbance strata (Appendix B). To better capture omission errors, which are more likely to occur near

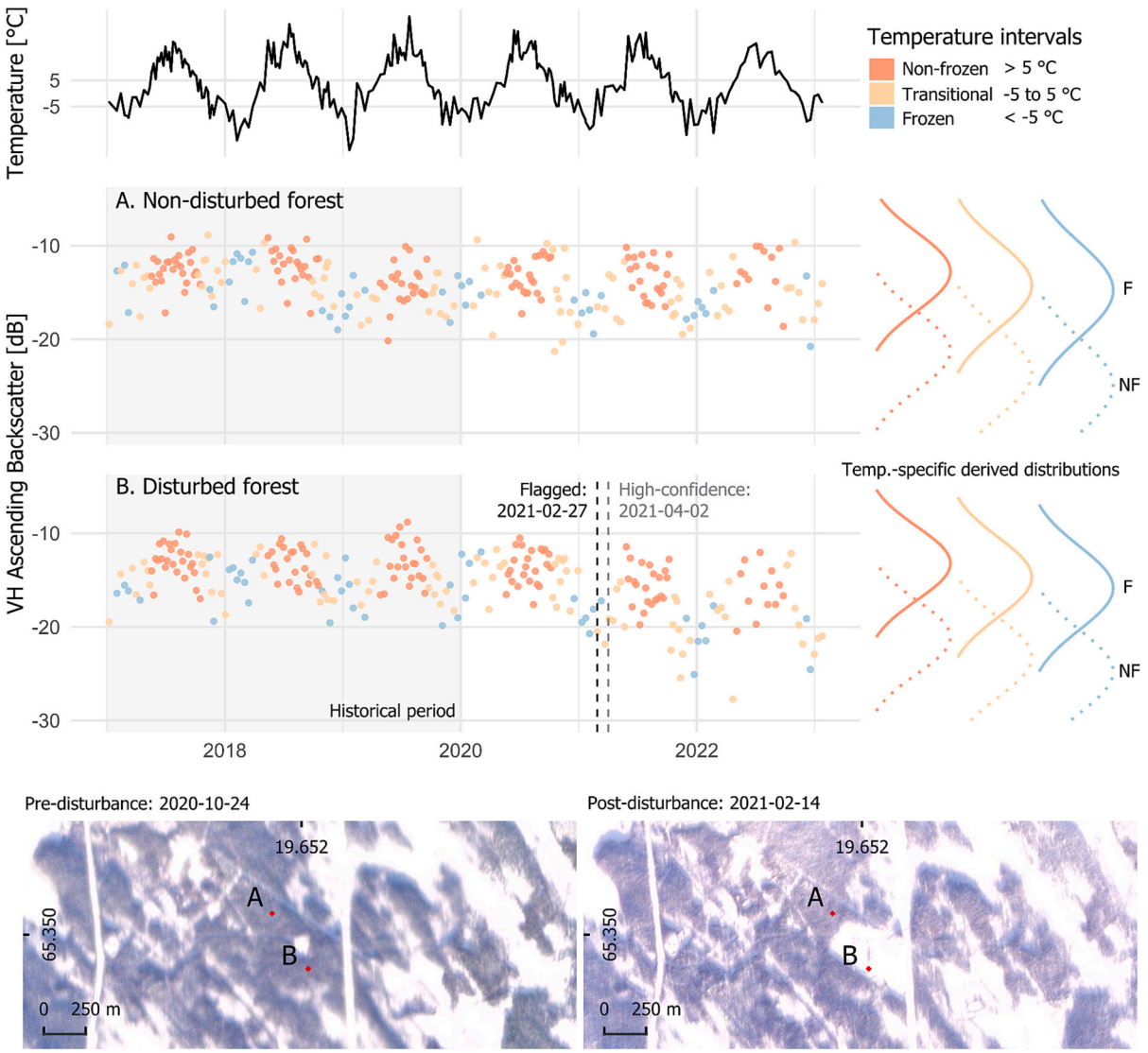


Fig. 3. Example of the disturbance detection methodology in Northern Sweden. The ERA5-Land temperature values at the time of observation are shown in the top time series plot. Below this are Sentinel-1 VH backscatter time series in ascending orbit for a non-disturbed pixel (A) and a disturbed pixel (B). The temperature-specific derived distributions of backscatter for forest (F, solid lines) and non-forest (NF, dotted lines) are depicted on the right and were derived from the observations in the historical period (in grey). The maps show the last available pre-disturbance optical image (left) and first available post-disturbance optical image (right) available from Planet Labs.

existing forest disturbances (Olofsson et al., 2020), one stable forest stratum was defined within a 200 m buffer surrounding detected disturbances (F < 200 m), while the second stratum included all areas outside of the buffer (F > 200 m). Both stable forest strata were derived from all pixels within the 2019 forest cover mask. To avoid bias towards large disturbance events, the two forest disturbance strata were defined by event size: the first included disturbances smaller than 2 ha (D < 2 ha), and the second included those larger than 2 ha (D > 2 ha).

We calculated potential sample sizes for a range of target standard errors ($se^T = 0.10, 0.25, 0.50\%$), with expected user accuracies (F < 200 m: 99%, F > 200 m: 99%, D < 2 ha: 90%, D > 2 ha: 90%) using Eq. (13) in Olofsson et al. (2020), resulting in sample sizes of 10,922, 1747 and 436 respectively. We chose to use 1000 samples ($se^T = 0.33\%$), split equally into 250 samples per stratum similar to previous disturbance mapping efforts (Balling et al., 2023; Hansen et al., 2016, 2013; Reiche et al., 2021; Turubanova et al., 2023) while taking sampling costs into consideration. Both the sampling and population unit corresponded to a 10 m × 10 m Sentinel-1 pixel. Disturbance patch boundary-related ambiguities related to partial tree cover, radar geometry and location inaccuracies were not regarded as errors in order to avoid penalizing the alert system, see Appendix C for full details.

The accuracy of forest disturbance detection was estimated while correcting for unequal inclusion probabilities between strata, reflecting the disproportionate allocation of sample points to strata areas (Stehman et al., 2003). The 2019 forest cover mask and the mapped forest disturbances were used to generate the strata samples and calculate area statistics and inclusion probabilities for each stratum (Stehman et al., 2003). A weighted confusion matrix was created using estimation weights derived from the inclusion probabilities (Appendix B), and area-adjusted user accuracy (UA; 100-commission error), producer accuracy (PA; 100-omission error) and F1 (2 * (PA * UA) / (PA + UA)) were calculated (Stehman, 2014). To avoid penalizing our disturbance detection methodology for remaining errors in the forest cover mask (Fig. 1), we also calculated accuracies separately after excluding 45 affected validation samples in the disturbance strata (Table 1).

UA, PA and F1 were calculated for entire Europe, as well as separately for the four defined geographic regions: North, East, South, and West Europe (Fig. 1). Post-stratification was applied and we recalculated

Table 1
Estimated overall and region-specific user accuracy (UA: 100-comission error) and producer accuracy (PA: 100-omission error) and F1 scores for all high-confidence alerts, with and without the exclusion of forest cover mask errors [%] (± standard errors [%]). Samples [n] overall and per region.

	All detected disturbances				Excluding forest cover mask errors			
	UA	PA	F1	n	UA	PA	F1	n
Overall	91.2 (1.3)	74.5 (6.0)	82.0	1000	99.1 (0.4)	76.0 (5.8)	86.0	955
North	94.7 (1.5)	75.9 (8.3)	84.2	407	100 (0.0)	76.8 (8.1)	86.9	393
East	88.0 (3.3)	82.6 (14.4)	85.2	190	97.8 (1.5)	84.1 (13.4)	90.4	179
South	90.7 (3.4)	51.7 (14.0)	65.9	167	98.6 (1.4)	53.8 (13.9)	69.6	161
West	85.7 (3.3)	84.6 (13.0)	85.2	236	98.0 (1.4)	86.3 (11.8)	91.8	222

the inclusion probabilities based on the updated area statistics (Appendix B) and sample sizes (Table 1) for each region (Tsendbazar et al., 2021).

4.2.2. Assessing detection timeliness

We assessed the detection timeliness by measuring the delay between the detection date of high-confidence alert and a reference date, visually determined as the first day Planet daily imagery showed disturbance with canopy cover loss greater than 50% within the Sentinel-1 pixel area. The disturbance timeliness is therefore relative to the earliest available optical image of disturbance, rather than the true date. We excluded samples where the time difference between the last visible stable forest phase and the first observed disturbance exceeded 31 days (80 samples removed) to minimize large uncertainties in the reference date. The resulting median time between pre- and post-disturbance Planet image pairs was 6.5 days. Only valid disturbance samples were included, excluding commission and omission errors as well as errors related to the forest cover mask. We reported the median, 25th and 75th percentile for all remaining samples ($n = 370$) across Europe and within each of the four validation regions (Fig. 1).

We also assessed the time required to move from low- to high-confidence alerts (Reiche et al., 2018b). We reported the median delay and the percentage of alerts moved from low- to high-confidence from 36 days (the 30-day minimum confirmation window plus one observation) to 90 days after first detection, using the average of the backscatter-based and GLCM-based values.

4.3. Comparison with annual optical disturbance products

We compared the mapped area of high-confidence disturbance alerts with the Global Forest Change (GFC) (Hansen et al., 2013) and European Forest Disturbance Atlas (EFDA) (Viana-Soto and Senf, 2025) products for the 4-year 2020–2023 period. This allows for the evaluation of relative differences in the spatial and temporal trends, but it does not provide an unbiased comparison of the true area of disturbances. Both GFC and EFDA are Landsat-based and provide annual forest disturbance estimates for entire Europe with a 30 m pixel size. Hansen et al. (2013) do not use a minimum mapping unit (MMU), while Viana-Soto and Senf (2025) use a three pixel (0.27 ha) MMU. Two comparisons were performed: First, we resampled the optical products to 10 m pixel spacing and masking them using the same forest cover mask as this study in order to mitigate differences in product resolutions and forest cover definitions. Second, we used the native resolution of each product and their original forest masks. For both comparisons, we summarized the disturbed area and number of disturbance events using 4-connected pixels for very small (≤0.27 ha; 3 Landsat pixels), small (0.27–1 ha), medium (1–5 ha) and large (≥ 5 ha) disturbance events. The area estimates produced are based on simple pixel counts and not corrected for commission and omission bias.

4.4. Analyzing seasonal European disturbance regimes

We investigated disturbance seasonality for the 2020–2023 period by performing an additional postprocessing step on high-confidence alerts to correct for late detection bias, i.e. a systematic shift between the detected disturbance date and the reference optically-derived date. Late detection bias is related to remaining canopy structure in the core of large disturbance events which reduces contrast between forest and non-forest in the C-band radar signal (Balling et al., 2024, 2023; Mitchell et al., 2014). During autumn, large fluctuations in temperature and high precipitation levels lead to high variability in historical forest conditions, while in winter, freezing conditions lower overall backscatter. Disturbance event edges are typically rapidly detected due to radar shadowing effects (Bouvet et al., 2018), which cause a strong drop in backscatter year-round.

We leveraged this by applying event-based bias correction to adjust disturbance dates in the event core using earlier dates at event edges. We adapted the method used by Kennedy et al. (2012). First, we defined the bias-corrected detection date as the 5th percentile of all detection dates within each four-connected cluster. This threshold was chosen instead of the earliest detection date to reduce the influence of false early detections. Second, bias-corrected dates were assigned only to clustered detections occurring within a 275-day temporal window (~9 months); these were considered as discrete events. The underlying assumption is that spatially connected but temporally separate events occur at relatively fixed times of the year, such as such as large fires during the dry periods or forest management in winter (Sebald et al., 2021; Wohlgemuth et al., 2022). The choice of 9 months rather than one year allows for some variation in disturbance seasonality between years. For example, this prevents directly adjacent clearcuts occurring in January and December of the same year from both being attributed the bias-corrected date of the earliest clearcut. Finally, the process was repeated until no further improvement was found (three times), in which each iteration was applied only to the remaining uncorrected pixels. We reported the improvement in timeliness compared to the timeliness of the near-real time monitoring scenario in Section 4.2.

5. Results

5.1. European forest disturbance alerts

We generated weekly European disturbance alerts from 2020 to 2023 (Fig. 4). The product demonstrates the ability to detect a wide range of spatiotemporally varying disturbances across Europe. Examples include clearings following bark beetle damage in South Tyrol, Italy (A), intensive large-scale clearcuts in Rapla, Estonia (B), harvesting (in 2020 and 2021) followed by a large fire (2022) in Landes, France (C) and small-scale group fellings in Suceava, Romania (D). The disturbance alerts highlight localized areas with high relative disturbance pressure (~7.5% to ~30%), including large fire events in Portugal, Sicily, and Greece; bark beetle damage in Germany and Czechia; and harvesting in France, Scotland, Hungary, Serbia, and Romania. In addition, wide areas of moderate disturbance pressure (~2.5% to 7.5%), primarily related to forest management practices, were observed in Poland, Ireland, and across northern Europe.

Performance was improved compared to the baseline method developed for the tropics (Reiche et al., 2021), demonstrating the advantage of integrating temperature and forest type data into the detection method (Fig. 5). Commission errors were in reduced in northern Sweden related to freezing temperatures (A) and in Southern Italy related to shifts in seasonal leaf phenology (B). An example is shown in (C) of simultaneous reduction of commission error and improvement of detection timeliness in Estonia, where the clearcut areas could be more reliably attributed to late 2020 than early 2021.

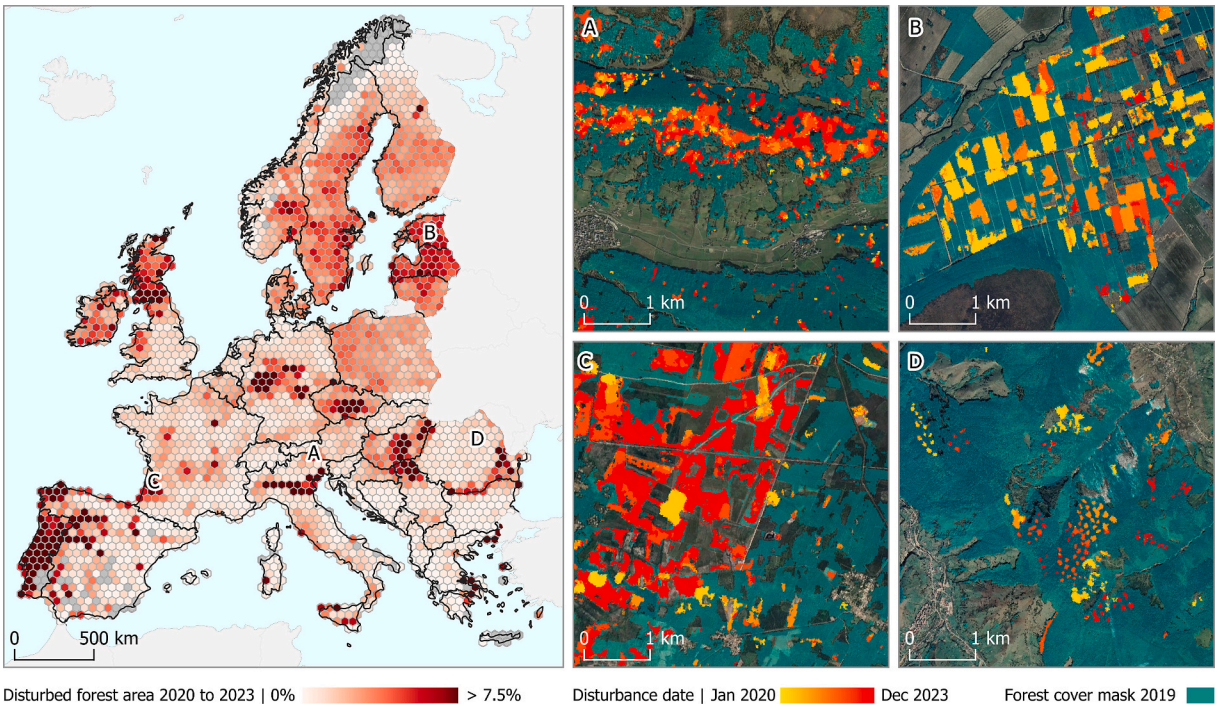


Fig. 4. Detected forest disturbances from January 2020 to December 2023, summarized as percentage of disturbed forest area within ~50 km hexagonal grid cells. Insert maps show disturbance alerts within the 2019 forest cover mask for four regions with distinct disturbance regimes (Google Earth imagery used as basemap): (A) Bark beetle damage and clearing in South Tyrol, Italy, (B) clearcuts in Rapla, Estonia, (C) mixed harvesting and fire in Landes, France, and (D) small-scale group fellings in Suceava, Romania.

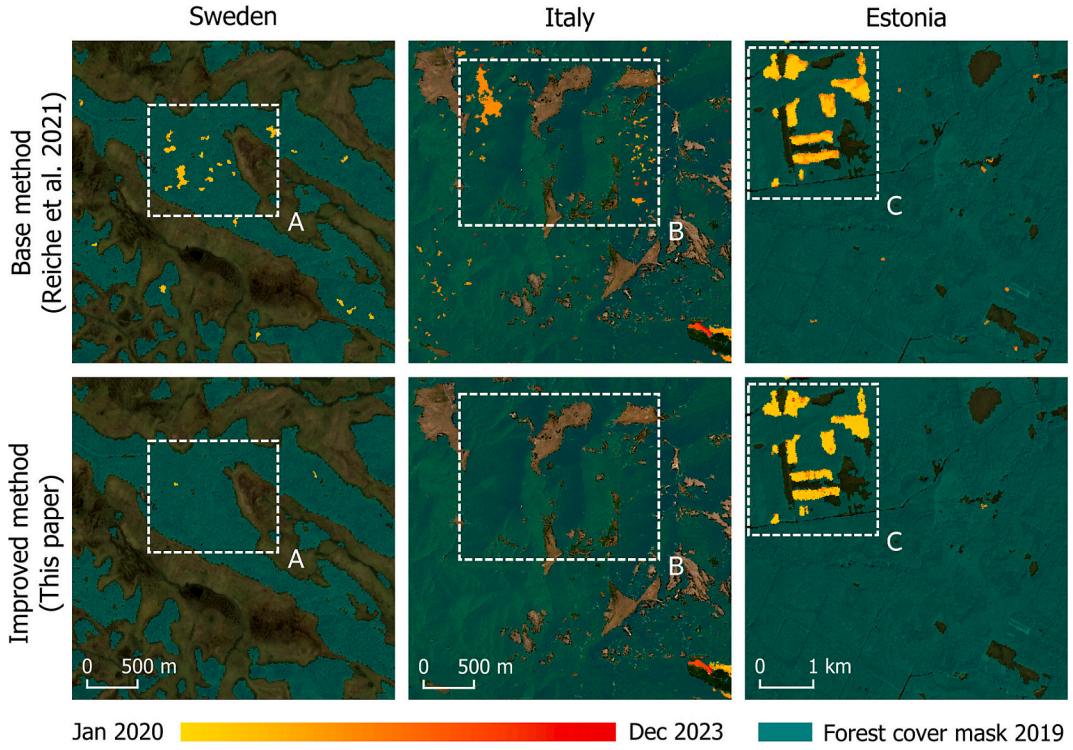


Fig. 5. Comparison between the baseline method used in the humid tropics (Reiche et al., 2021) and the improved method developed for Europe (this paper). Alert dates are shown within the 2019 forest cover mask for three example locations (Planet-Labs imagery used as base maps): (A) reduction of commission error in sparse peatland conifer forest in northern Sweden, (B) in dense deciduous forest in southern Italy, and (C) improvement of detection timeliness in dense conifer forest in Estonia.

5.2. Spatial accuracy and timeliness

Spatial validation of forest disturbance alerts revealed high overall accuracies (Table 1). The user accuracy (UA; 100 - commission error) was 91.2% (±1.3%), while producer accuracy (PA; 100 – omission error) reached 74.5% (±6.0%). The highest UA of 94.7% (±1.5%) was found in northern Europe. The PA was similar across all regions (ranging from 75% to 84.6%) with exception of southern Europe, where PA was low: 51.7% (±14.0%). This lower PA in southern Europe was driven by just 3 forest samples classified as omission error (Appendix B). In total, ten stable forest samples were classified as omission errors, all stemming from the within-200 m buffer stratum (Appendix B). With forest disturbances associated with remaining forest cover mask errors excluded, overall UA improved to 99.1% (±0.4%) with consistently high accuracies across all regions, resulting in near-zero commission error across all regions (ranging from 97.8% to 100%). PA also improved slightly, reaching 76.0% (±5.8%) across all regions.

The detection timeliness of the disturbance alerts showed an overall median detection delay of 27 days (p75 = 109; p25 = 5) (Fig. 6). The shortest median delay occurred in spring (38 to 14 days) and summer (23 to 10 days), while longer delays were observed during winter (57 to 20 days), particularly in northern Europe. Autumn detections exhibited the longest delays and the highest variability (156 to 24 days), especially in northern and western Europe.

Applying event-based bias-correction substantially reduced late detection bias. The overall median detection delay across all seasons and regions was reduced to 1 day (p75 = 17; p25 = −17.5). The greatest improvements were observed in winter (9 to −8 days) and autumn (10 to −9 days). However, some negative detection delays were introduced, particularly in eastern and western Europe during summer (−27 and −11 days, respectively).

The analysis of the time delay between issuance of low- and high-confidence alerts revealed that 66% of low-confidence alerts achieved high-confidence within 36 days, 80% within 48 days, and 95% within 90 days.

5.3. Comparison with annual optical disturbance products

Our method resulted in a greater detection capability of small disturbances (≤0.27 ha, equaling 3 Landsat pixels), but a lower total detected disturbed area in the 4-year 2020–2023 period compared to the optical-based products within our 2019 forest cover mask. We detected 332 kha (2.1 M events) in the ≤0.27 ha class compared to 253 kha (1.6 M) and 207 kha (1.2 M) for Viana-Soto & Senf and Hansen et al. respectively (Fig. 7). The same increased capacity to detect small disturbances was found using the native resolutions and original forest masks of the optically-based products (Appendix D.1 and D.2). Fig. 8A depicts a visual example from Romania, in which the high spatial detail allowed for the detection of small-scale group felling pattern with a clear delineation of disturbance patch borders. However, we found a lower total disturbed area of 3174 kha compared to 5037 kha in Viana-Soto and Senf (2025) and 5290 kha in Hansen et al. (2013) (Fig. 7). This was primarily driven by missed detections in the large (≥ 5 ha) disturbance class, in which we detected 985 kha compared to 2079 kha and 2461 kha for Viana-Soto & Senf and Hansen et al. respectively (Fig. 7), related to large and low-intensity disturbances such as thinning (Fig. 8B) and fire

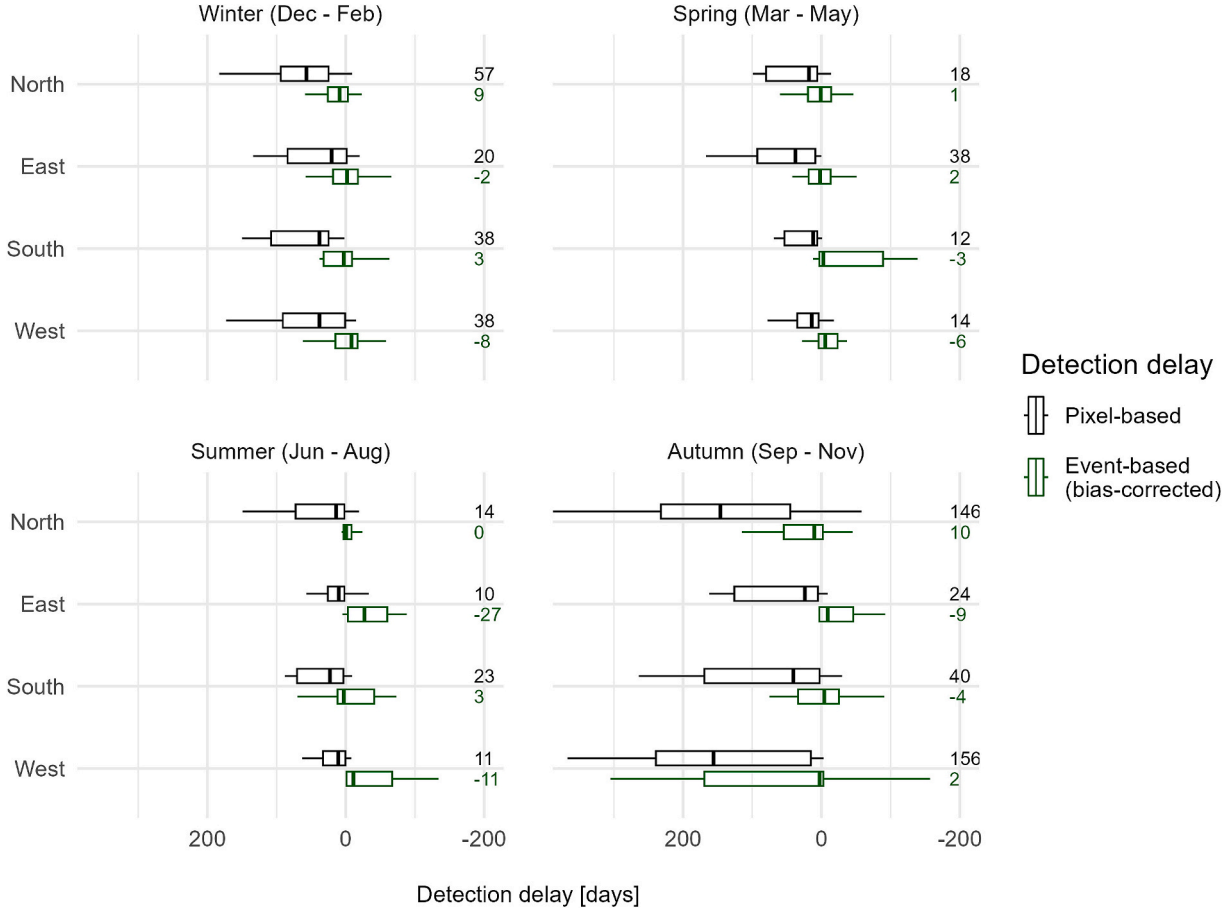


Fig. 6. Detection delay (in days) relative to the first visible disturbance in optical (Planet Labs) imagery, shown per geographic region and three-month season. Delays are reported for both pixel-based (using the near-real time alerting product) and event-based (using bias correction) results. Positive values indicate detection later than optical imagery, while negative values indicate earlier detection.

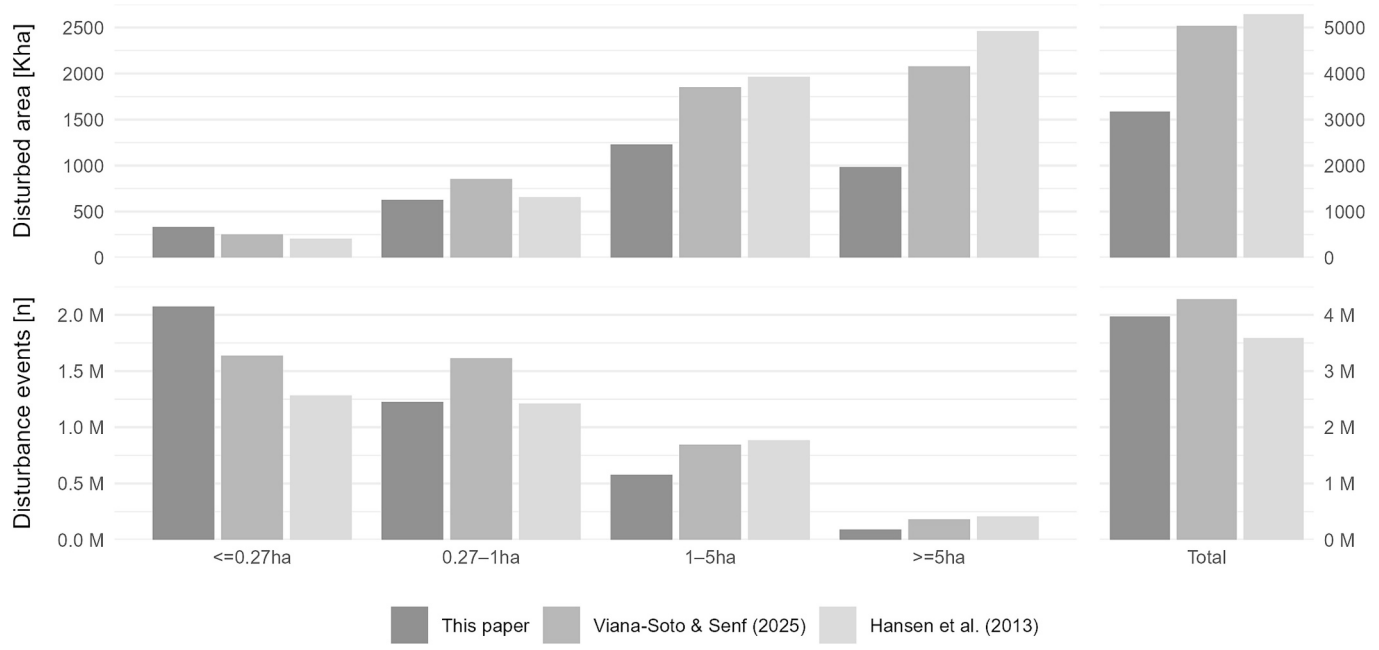


Fig. 7. Disturbed area in Kha (top) and disturbance event counts (bottom) by disturbance size class comparison for the 2020–2023 period, comparing results from this study with two Landsat-based annual disturbance products: Viana-Soto and Senf (2025) and Hansen et al. (2013), masked using the forest cover mask in this study. The smallest size class (≤0.27 ha) represents disturbances equal to or smaller than three 30 m Landsat pixels, the minimum mapping unit used in Viana-Soto & Senf. Note the different scales for the left and right plots.

(Fig. 8C).

5.4. Seasonal patterns in European disturbance regimes

We uncovered large seasonal variation in the timing and spatial distribution of disturbances across Europe on a ~ 50 km hexagonal grid (Fig. 9). Disturbance regimes often showed one or two recurring seasonal peaks, revealing distinct spring-, winter-, or summer-dominated patterns. Winter-dominated disturbance regimes were prominent in parts of central and eastern Europe, as shown in Fig. 9A, which depicts harvesting activity in northern Finland with secondary peaks in spring and summer. High spring disturbance pressures (>40% of total disturbance area for the 4-year 2020–2023 period) were observed across western Europe, Italy, the British Isles and southern Scandinavia. For example, Fig. 9B illustrates an early-spring peak associated with forest management in southern Belgium. Summer disturbances were most concentrated in Sicily, Greece and the Balkan. Fig. 9D shows an example from southern Serbia, where harvesting activity is centered around the summer months. In contrast, some areas lacked a clear seasonal pattern due to irregular disturbance timing. This was evident in regions with low forest cover and fire-driven disturbance regimes, such as parts of Spain, Portugal, Italy and Greece. Fig. 9E depicts an extreme case from Portugal, where a single wildfire in summer 2022 accounted for 20% of the total disturbed area over the entire 4-year period. Areas like alpine Switzerland (Fig. 9C) exhibited a relatively uniform distribution of disturbance events, showing no distinct seasonal pattern.

6. Discussion

Detecting forest disturbances at high temporal and spatial resolution is important for our understanding of the mixed anthropogenic and natural disturbance regimes of Europe, as well as for supporting law enforcement agencies and policymakers (Senf and Seidl, 2021; Woodcock et al., 2020). In this study we presented a novel adaptation to a Sentinel-1 based tropical forest monitoring method (Reiche et al., 2021), which has allowed us to produce the first year-round European radar-based forest disturbance alerting product.

The high spatial detail of the Sentinel-1 system allowed for the detection of small disturbances and accurate determination of disturbance event borders. This methodological framework is scalable and applicable to operational settings, enabling the provision of near-real time disturbance alerts similar to those currently available for the tropics (Reiche et al., 2018a). Consistent with previous alerting products (Hansen et al., 2016; Reiche et al., 2021), our alerts provide timely detection of new forest disturbances without quantifying their exact area, but enable the assessment of relative spatial and temporal trends.

6.1. Accurate and consistent forest disturbance alerting

Our radar-based forest disturbance alerting method demonstrated high accuracy, characterized by a user accuracy (UA) of 91.2% (±1.3%) and a producer accuracy (PA) of 74.5% (±6.0%). When excluding errors associated with local inaccuracies of the European-scale forest cover mask applied here (e.g. overestimation of forest height and density), the UA improved to 99.1% (±1.3%). This highlights the benefits of using refined national or local-scale forest cover products tailored to specific user needs.

The relatively low PA indicates a conservative approach that prioritizes minimizing commission errors, which is essential for reliable alerting systems (Reiche et al., 2024, 2021; Woodcock et al., 2020). Some commission errors persisted, primarily resulting from local inaccuracies in the underlying forest cover mask (Fig. 1). Particularly in areas with sparse or low-canopy forests such as alpine and Mediterranean regions, backscatter variability due to soil moisture and snow cover fluctuations was more pronounced and difficult to mitigate solely with temperature adjustments (Pulliainen et al., 1996). Some omission errors also remained, especially in the fire-driven disturbance regime of Southern Europe (PA = 51.7%). Out of three omission errors in Southern Europe, two were events larger than 2 ha, suggesting the sensitivity of the alerting methodology to large, low intensity fires is relatively low. C-band radar is not only sensitive to structural changes in the canopy (van der Woude et al., 2024), but also to post-disturbance moisture fluctuations and remaining structure which have been shown to affect detection of large disturbances such as fires (Balling et al., 2023, 2021; Bouvet

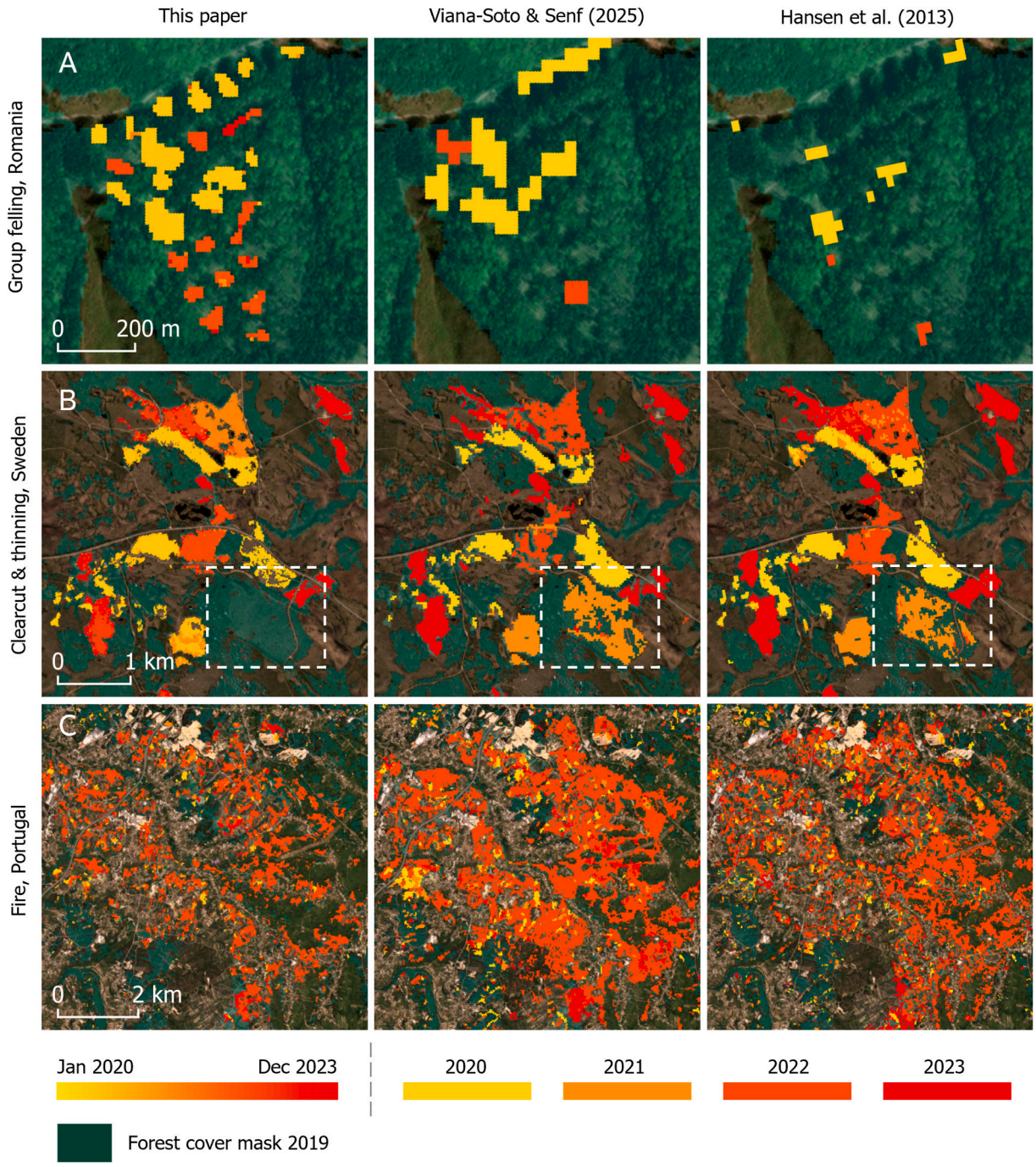


Fig. 8. Visual comparison at different scales between the forest disturbance product from this study, Viana-Soto and Senf (2025), and Hansen et al. (2013), for (A) small-scale group felling in Romania, (B) clearcuts and thinning in northern Sweden (Note: thinning — highlighted in white — was omitted in our method because the Sentinel-1C-band radar signal is not sensitive to low-intensity forest disturbances) and (C) large-scale fire in Portugal. No additional forest cover masking was applied. The forest cover mask for 2019 is shown in green. A May 2024 Planet Labs composite image is used as a basemap. Note that the scale used for this paper is continuous, while the annual products are shown with discrete scales. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al., 2018; Gibson et al., 2023).

Compared to other radar-based disturbance products, our method displayed roughly similar spatial accuracies. Mermoz et al. (2024) achieved similarly high UA (99.4%) and PA (80.9%) for clearcuts in France, while Mullissa et al. (2024) reported lower accuracies (UA of 78%, PA of 40%) across Europe. A simple analysis of disturbed area between Mermoz et al. (2024) and this study in France revealed that our method detected ~26% lower disturbed area within our 2019 forest cover mask (Appendix E). This indicates Mermoz et al. likely exhibits lower omission errors in France, in line with expectations for a locally-parameterized model. However, both removed observations in the winter due to freezing temperatures (Mullissa et al., 2024) and/or snow cover (Mermoz et al., 2024; Mullissa et al., 2024). This can lead to late or non-existent winter detections, especially in areas which regularly experience these conditions. Our methodology addressed these limitations by allowing the detections to continue uninterrupted despite seasonal and daily weather-induced fluctuations.

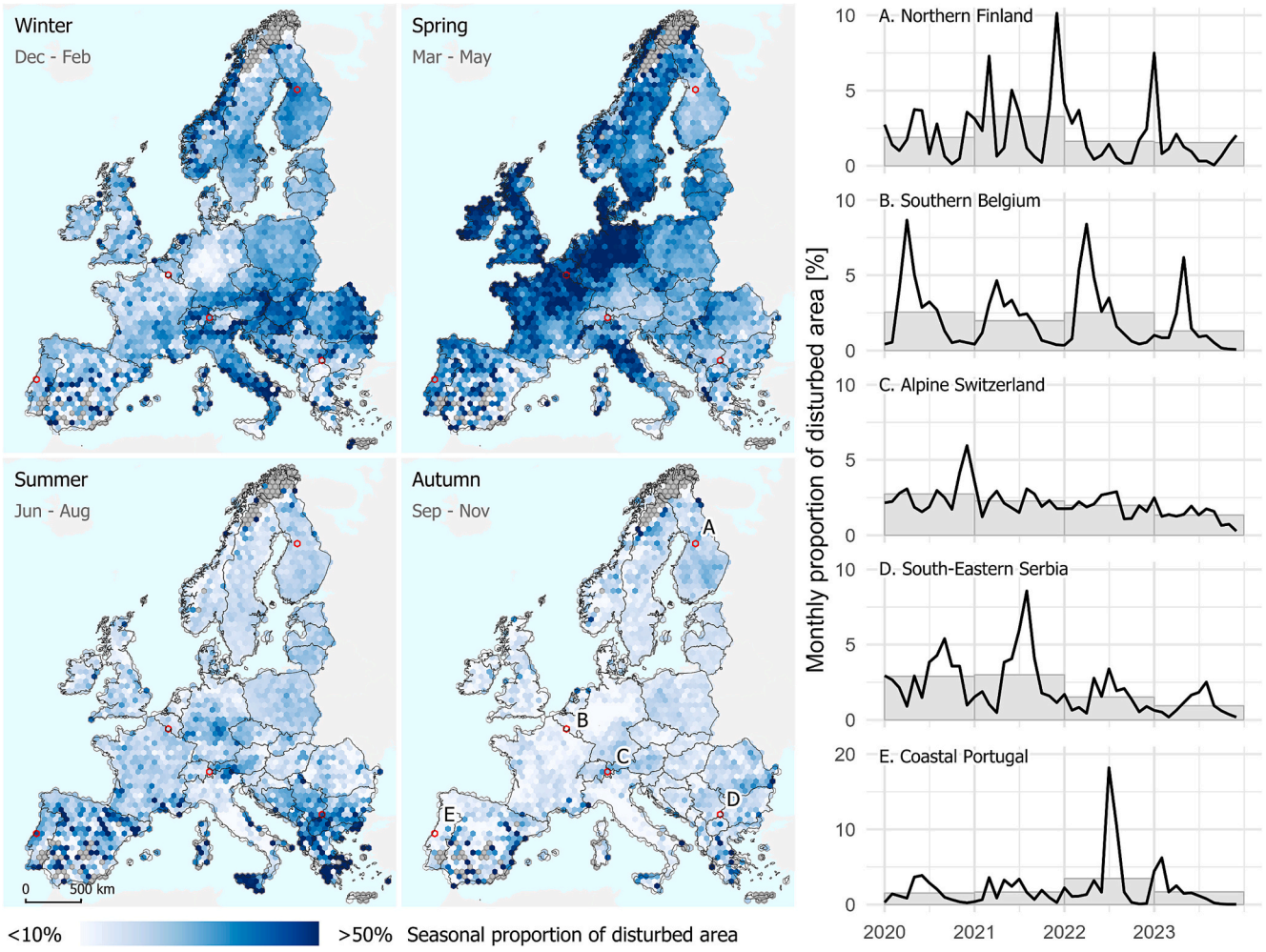


Fig. 9. Seasonal proportion of disturbed area in a ~ 50 km hexagonal grid. Five selected grid cells (A-E) are highlighted in red on the map and correspond to time series plots of monthly proportion of disturbed area per cell with a three-month moving window applied. Grey bars in the background indicate the monthly values per year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6.2. Detection timeliness

We achieved a low overall detection delay of 27 days relative to the earliest visible disturbance in Planet imagery, with particularly good performance in spring and summer (Fig. 5). In some cases, negative detection delays were found, indicating earlier detection than the first optical reference image in which a disturbance was visible. This was possible because detections can occur in the time period between pre- and post-disturbance optical image pairs (median of 6.5 days). A greater detection delay was found in autumn and winter months in northern and western Europe, related to remaining structure after disturbance events and climatic effects. In addition, natural disturbances such as bark beetle, wind, and fire events are known to often result in minimal decreases in backscatter (Balling et al., 2021; Hechtl et al., 2025; Tanase et al., 2018), or even increased backscatter (Rüetschi et al., 2019). The detection timeliness of natural disturbances is therefore likely lower than indicated in the overall assessment of timeliness, as these events are often only fully detected when post-disturbance salvage logging is performed.

Analysis of disturbance seasonality benefited from bias-corrected, event-based assessment to overcome late detections in autumn and winter. Applying event-based late detection bias corrections (Kennedy et al., 2012) significantly reduced the overall median late detection bias to 1 day relative to the earliest available optical image. However, this correction introduced minor early detection bias in some cases, particularly during summer months in Eastern Europe. We found that this overcorrection was mostly driven by samples located in bark beetle-affected disturbance patches in Czechia which spanned very large areas, and were therefore more prone to being assigned excessively early detection dates in some areas. This suggests further refinement of the simple event-based timing correction method could reduce early detection bias in large multi-event disturbance events.

6.3. Comparison with annual optical disturbance products

The map comparison between our radar-based alerting method and annual optical-based disturbance products revealed the greater spatial detail of our product compared to the Global Forest Loss (GFC) and European Forest Disturbance Atlas (EFDA) products, visible in a greater detected disturbance area and number of detections for fine-scale ($\leq$ 0.27 ha) disturbances (Fig. 7). This indicates that small forest disturbance events are not well-detected in available Landsat-based monitoring products (Hansen et al., 2013; Viana-Soto and Senf, 2025). Our overall detected area was significantly lower, as the relative disturbance area detected in our product decreased with disturbance size compared to the optical products (Fig. 7). The large omission in the $\geq$ 5 ha class compared to the optical products (Viana-Soto and Senf, 2024) likely stems from the lower sensitivity of the Sentinel-1C-band backscatter signal to detect low intensity non-stand replacing forest disturbances (Balling et al., 2023, 2021; van der Woude et al., 2024).

While a direct comparison of accuracy figures is challenging due to differences in validation strategies, forest masks, and spatial & temporal scales, the general pattern is consistent with our findings: our method exhibited fewer commission errors and more omission errors than existing annual optical-based forest disturbance products, while providing higher spatial and temporal detail. Viana-Soto and Senf (2025) reported 89.4% UA and 79.5% PA for detected disturbances in the period 2001–2023. Turubanova et al. (2023) reported 81.7% (±1.9) UA and 95.0% (±1.9) PA for their tree canopy removal product, and 59.5% (±4.3) UA and 71.4% (±2.5) PA for the GFC product (Hansen et al., 2013) using the same validation samples for the 2001–2021 period. The high spatial detail and consistent year-round detection capabilities of radar-based monitoring and the sensitivity of optical products to large, low-intensity disturbances highlights their complementarity.

6.4. Seasonality in European disturbance regimes

The assessment of seasonality in European disturbance regimes demonstrated that disturbance regimes vary not only spatially, but also seasonally throughout Europe. The overall patterns found here appear to relate largely to the presence and degree of forest management rather than natural disturbances (Sebald et al., 2021; Wohlgemuth et al., 2022). The effect of varying forest management styles is strongly visible in differences in seasonal patterns along country borders, such as between Poland and Germany, or between Serbia, North Macedonia and Kosovo (Fig. 9). Large scale managed forestry in northern Europe is characterized by winter and late spring harvest, since heavy harvesting machinery requires either dry or frozen conditions to avoid causing damage to access roads and forest plots (Lehtonen et al., 2019). Sanitation felling of bark-beetle affected spruce stands often occurs in early spring to avoid further spreading (Hlásny et al., 2019), as we found to be the case in western and central Europe. Summer fires are known to be common in Greece and the Balkan peninsula, while fires in the interior of Spain also occur independent of season (Galizia et al., 2021). Remote and inaccessible forests like those in alpine Switzerland show relatively little variation in seasonality, as they are affected by a wide range of natural disturbances throughout the year (Bebi et al., 2017). Only a 4-year period was analyzed, so these patterns were likely driven by large environmental and societal events, such as the 2018–2020 droughts which led to widespread bark beetle outbreaks in central Europe (Thonfeld et al., 2022) and the 2019–2023 Covid-19 pandemic (Hlaváčková et al., 2024; Kronholm, 2023).

6.5. Implications and outlook

The radar-based forest disturbance alerting methodology presented here expands our ability to monitor European forests by providing alerts with enhanced temporal and spatial detail and with a low data delivery lag compared to annual products. This has implications for forest management and conservation strategies, aligning their efforts closely with objectives outlined in the European 2030 Biodiversity Strategy and the 2030 Forest Strategy (European Commission, 2021b, 2021a). The rapid provisioning of disturbance data, in combination with local expert knowledge, provides actionable information that can strengthen law enforcement capabilities in detecting and responding to illegal logging (Thompson and Magrath, 2021), and refine national forest inventories (Coops et al., 2022). The high temporal detail can enable further studies into intra-annual dynamics of disturbance events, which are not possible with annual-scale monitoring approaches and multi-annual national forest inventory cycles (Senf and Seidl, 2018; Wohlgemuth et al., 2022). In addition, the ability to identify fine-scale patterns in time and space could lead to better classification of disturbance agents and their impacts (Belenguer-Plomer et al., 2019; Rüetschi et al., 2019; Slagter et al., 2023).

There are many opportunities to further improve near-real time forest monitoring. The operationalization of Sentinel-1C and Sentinel-1D and the launch of Sentinel Next Generation in 2033 will ensure the consistent and rapid availability of data beyond 2030. Special attention should be given to the varying impacts which different natural disturbances can have on C-band radar backscatter in order to improve detection accuracy and timeliness. Integrating radar-based disturbance detection with optical remote sensing has proven to be highly complementary (Doblas et al., 2023; Reiche et al., 2024) and could substantially improve the detection accuracy of disturbances and characterization of disturbance drivers (Sebald et al., 2021; Viana-Soto and Senf, 2025). L-band radar missions, such as the recently launched NISAR and upcoming ROSE-L, are expected to provide greater detection capabilities and reduced detection delay by means of their increased sensitivity to structural changes in the forest (Balling et al., 2024). This will mitigate some of the current limitations in the use of C-band radar posed by seasonal variations, freezing conditions, and post-disturbance remaining structure (Balling et al., 2024; Cohen et al., 2024, 2021).

Advancements in near-real time weather and climate data integration such as improved and high-resolution temperature, precipitation, and snow-cover datasets could further reduce commission errors as a result of signal variations due to drought, freezing temperatures and snowmelt. This could also reduce omission error and increase detection timeliness particularly in mountainous and low-density forests (Cohen et al., 2021; Koskinen et al., 2010). This will be especially crucial when expanding this detection methodology beyond the currently used forest cover mask to lower density forests where the increased ground contribution of the radar signal (Magagi et al., 2002) requires a better characterization of moisture-induced fluctuations such as drought and snow melt. The availability of global near-real time climate data calls for the expansion of this methodology to boreal, temperate and dry forests outside of Europe.

7. Conclusion

This study demonstrated a radar-based approach for near-real time monitoring of forest disturbances across Europe by integrating Sentinel-1 radar data with ERA5-Land temperature and Copernicus forest type datasets. The resulting alerts enhance geographic coverage and temporal consistency of forest disturbances compared to existing Europe-wide radar-based near-real time products and complement available annual Landsat-based products. These alerts offer timely forest disturbance information which can contribute to more effective forest management, biodiversity conservation, and more accurate carbon accounting, in particular when combined with expert knowledge on disturbance sources. The high temporal detail of the data allowed for the first year-round European-scale assessment of intra-annual disturbance seasonality. Additionally, combining radar-based systems with existing optical products will provide highly complementary insights, further enriching the understanding and management of forest disturbance dynamics. Future uptake of new L-band radar missions, improved climate data integration, and integration of monitoring systems represent immense opportunities for the enhancement of forest disturbance management.

CRediT authorship contribution statement

Sietse van der Woude: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Johannes Reiche:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Johannes Balling:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Gert-Jan Nabuurs:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Frank Sterck:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Anne-Juul Welsink:** Writing – review & editing, Methodology, Conceptualization. **Bart Slagter:** Writing – review &

editing, Methodology, Conceptualization. **Martin Herold:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Open AI's ChatGPT in order to aid in the generation of code for simple data summarization and visualization tasks, as well as improving readability, grammar and spelling. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

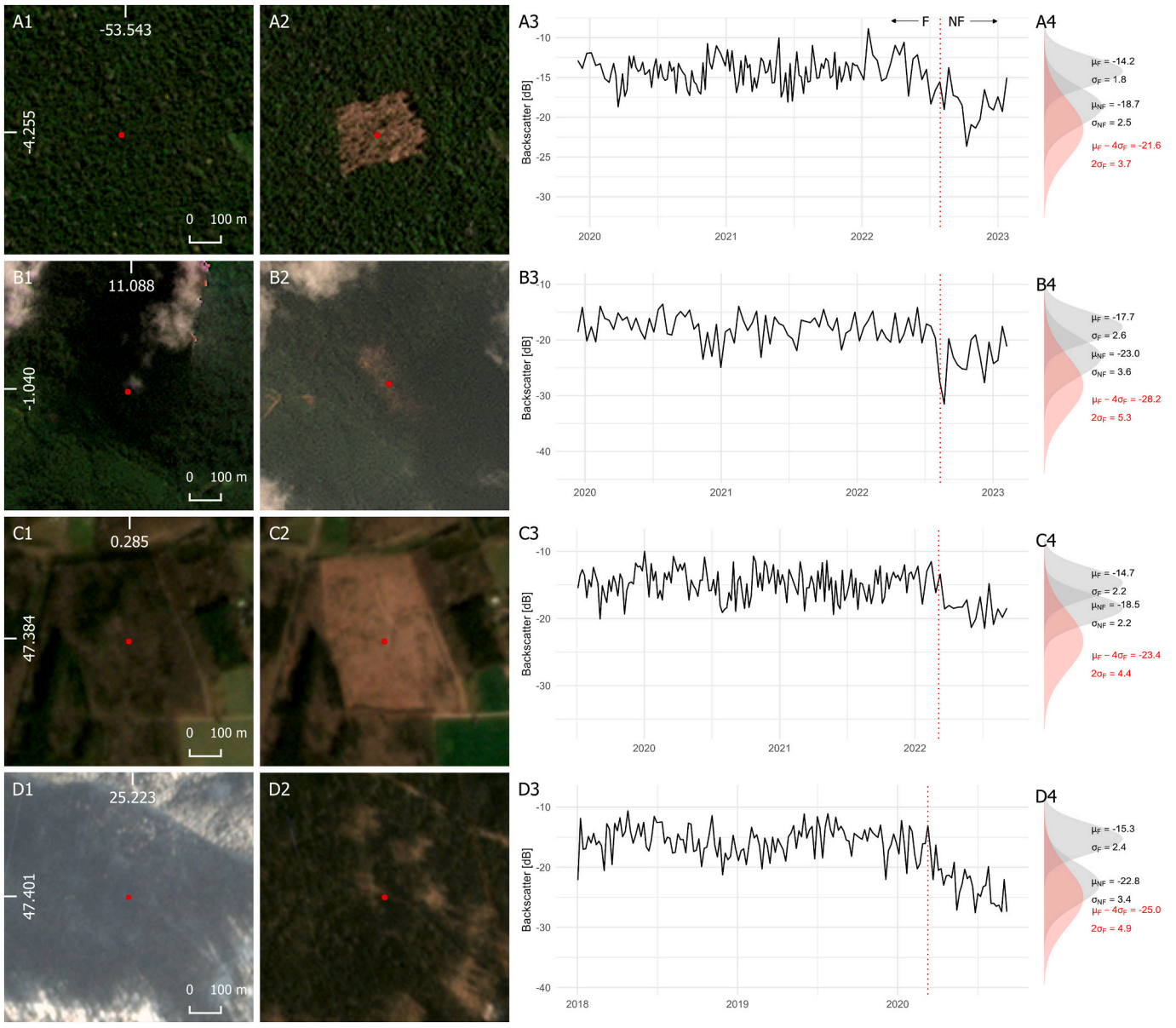


Fig. A.1. Time series and Sentinel-1 backscatter distribution examples from two locations (single S1 pixel) in the tropics detected by RADD (Reiche et al., 2021) (A: Brazil; B: Gabon) and two locations in Europe (C: France; D: Romania) detected as change in this paper. Maps A1-D1 depict stable forest (F) conditions, while maps A2-D2 depict non-forest (NF) conditions. In A3-D3 time series of VH backscatter for one orbit direction are shown. The vertical red line depicts the date of detection. A4-D4 show the measured F and NF backscatter distributions in grey, and the derived NF ($\tilde{x}$-4σ, 2σ) in red, which is based on the measured observations during F conditions. All imagery are Planet Labs monthly composites

In Fig. A.1, we depict several examples from the tropics and Europe of forest disturbances and the corresponding Sentinel-1 backscatter time series. We did not use the temperature intervals here to maintain comparability. In both the tropics and Europe, the measured NF values do not always reach the derived NF median (in red). However, due to the probabilistic nature of the detection method, a change can be detected given enough observations where the NF probability is greater than the F probability.

Appendix B

Table B.1
Overview of strata area proportions (Area [%]), correctly classified samples out of total samples per strata (Correct/Total) for four strata: stable forest within 200 m buffer around change (F < 200 m), stable forest outside 200 m buffer around change (F > 200 m), disturbance smaller than 2 ha (D < 2 ha) and disturbance larger than 2 ha (D > 2 ha). Also shown are commission (Com.) and omission (Om.) errors by region

		F < 200 m	F > 200 m	D < 2 ha	D > 2 ha	Com.	Om.
All regions	Area [%]	75.96	21.54	1.11	1.39	49	10
	Correct/Total	250/250	240/250	223/250	228/250		

(*continued on next page*)

Table B.1 (*continued*)

		F < 200 m	F > 200 m	D < 2 ha	D > 2 ha	Com.	Om.
Northern Europe	Area [%]	73.42	23.25	1.23	2.10	14	5
	Correct/Total	88/88	90/95	94/98	116/126		
Eastern Europe	Area [%]	74.59	23.13	1.13	1.15	12	1
	Correct/Total	40/40	54/55	44/53	39/42		
Southern Europe	Area [%]	84.57	13.93	0.80	0.70	7	3
	Correct/Total	65/65	30/33	36/42	26/27		
Western Europe	Area [%]	74.39	23.37	1.16	1.08	16	1
	Correct/Total	57/57	66/67	49/57	47/55		

Table B.2
See Table B.1., but excluding forest cover mask errors from the forest disturbance strata D < 2 ha and D > 2 ha.

		F < 200 m	F > 200 m	D < 2 ha	D > 2 ha	Com.	Om.
All regions	Area [%]	75.96	21.54	1.11	1.39	5	10
	Correct/Total	250/250	240/250	222/226	228/229		
Northern Europe	Area [%]	73.42	23.25	1.23	2.10	0	5
	Correct/Total	88/88	90/95	94/94	116/116		
Eastern Europe	Area [%]	74.59	23.13	1.13	1.15	2	1
	Correct/Total	40/40	54/55	43/45	39/39		
Southern Europe	Area [%]	84.57	13.93	0.80	0.70	1	3
	Correct/Total	65/65	30/33	36/37	26/26		
Western Europe	Area [%]	74.39	23.37	1.16	1.08	2	1
	Correct/Total	57/57	66/67	49/50	47/48		

Appendix C

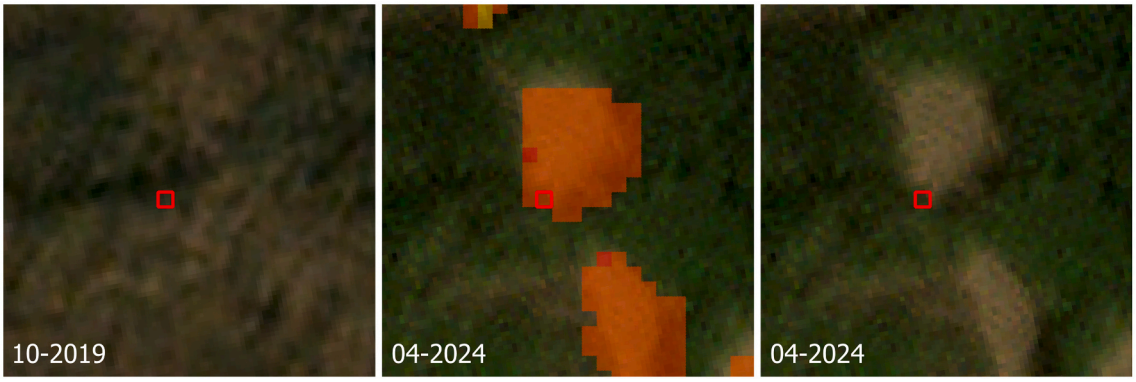


Fig. C.1. Example of a boundary pixel. The validation sample (in red) is located on the boundary of an alert patch (orange, semi-transparent). There is a clear disturbance visible in the Planet Imagery related to the alert patch, however it is difficult to unambiguously determine if the canopy at the location of the pixel was >50% removed

Boundary pixels typically reflect partial tree cover at the edges of larger disturbances, making validation challenging (Hansen et al., 2016), particularly with high-resolution Sentinel-1 data (Reiche et al., 2021). Besides the side-looking radar geometry, sub-optimal terrain correction in GEE could further introduce location inaccuracies (Flores-Anderson et al., 2023). Ambiguous pixels located on disturbance boundaries clearly visible in Planet imagery were labeled as 'boundary pixels' (Fig. C.1) rather than false detections (commission errors). In total 70 such cases were found. Similarly, disturbance events in the Planet imagery clearly related to alert pixels which appeared to be shifted due to side-looking radar geometry or location inaccuracies (Bouvet et al., 2018; Carstairs et al., 2022; Flores-Anderson et al., 2023), were labeled as 'edge case' rather than missed detections (omission errors), of which 3 were found. We did this to avoid penalizing the alert system, of which the main purpose is to detect disturbances rather than precisely estimate their area (Tang et al., 2019).

Appendix D

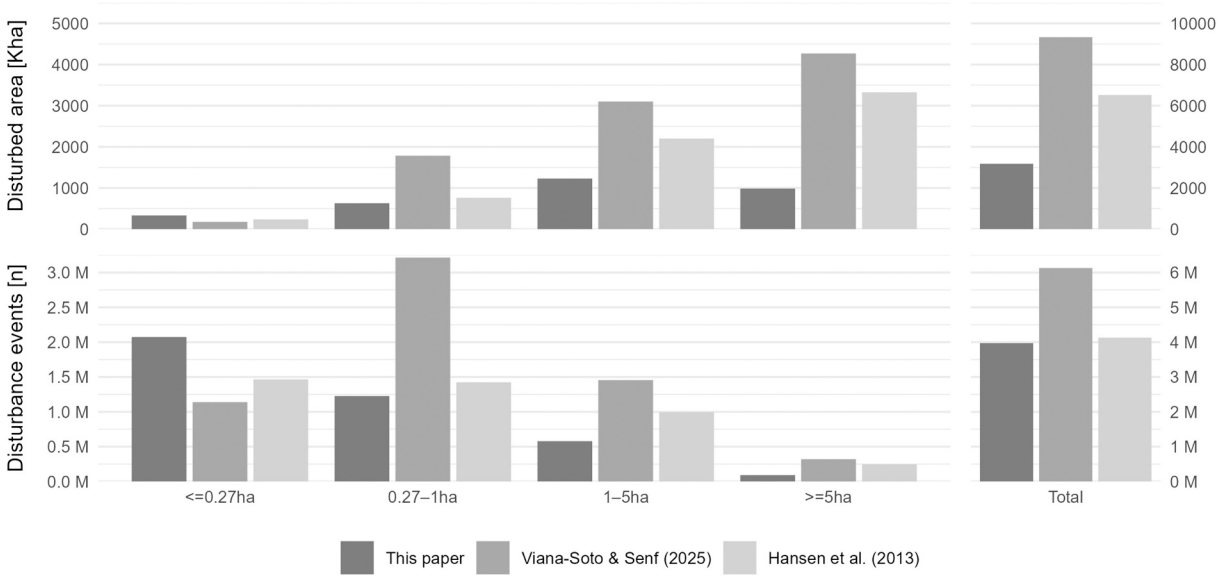


Fig. D.1. Disturbed area in Kha (top) and disturbance event counts (bottom) by disturbance size class comparison for the 2020–2023 period, comparing results from this study with Viana-Soto and Senf (2025) and Hansen et al. (2013) using each product's original forest mask. The smallest size class (≤0.27 ha) represents disturbances equal to or smaller than three 30 m Landsat pixels, the minimum mapping unit used in Viana-Soto & Senf. Note the different scales for the left and right plots

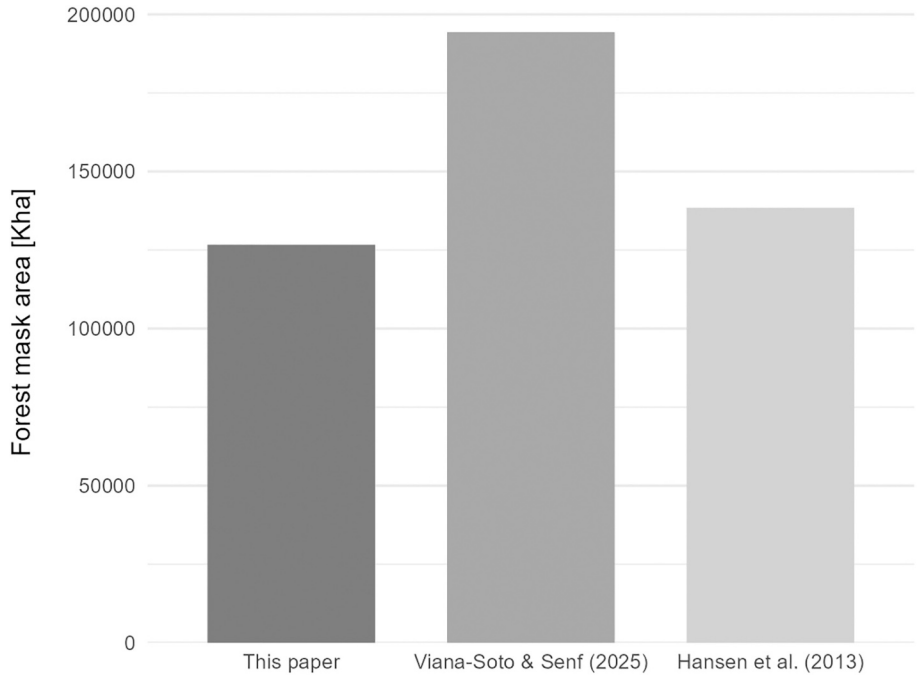


Fig. D.2. Total area in Kha covered by the forest masks used in this paper, Viana-Soto and Senf (2025), and Hansen et al. (2013). Note: areas of cover loss as detected by Hansen et al. pre-2020 were removed from the Hansen et al. forest cover mask. This was not done for Viana-Soto & Senf, since it is an estimate of forest land use, rather than forest cover.

Appendix E

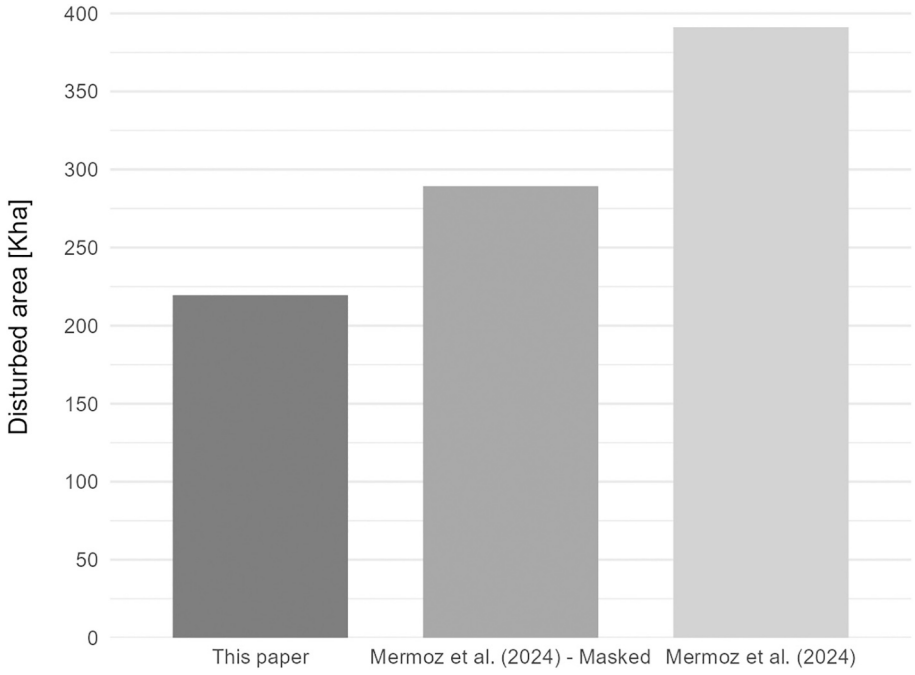


Fig. E.1. Disturbed area for France in Kha for the 2020–2023 period in this paper, Mermoz et al. (2024) masked using our 2019 forest cover mask, and Mermoz et al. (2024). Our 2019 forest cover mask yielded a 2020 forest area of 13,437 Kha in France, compared to 17,017 Kha in the forest cover mask of Mermoz et al. (2024) (after removal of pre-2020 detected areas).

Data availability

Data will be made available on request.

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